Five Formalizations of Proximal On-Policy Distillation for Flow Models

From deterministic field matching to stochastic transition and marginal mixtures

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Abstract

Flow models admit both deterministic ODE sampling and stochastic SDE sampling. A proximal distillation rule built from Gaussian transition likelihoods is therefore exact only for the SDE case; its deterministic limit is singular. This note develops five concrete arms under one flow-matching notation. Arm A0 is direct ODE teacher-field regression. Arm A1 adds an explicit field-space trust region with a local one-step transport interpretation. Arm A2 constructs a teacher gate from forward flow-matching prediction risk; it is solver-agnostic but is a predictive surrogate rather than a state-density ratio. Arm A3 is the existing SDE transition-mixture P-OPD, whose detached-responsibility loss matches the exact mixture-KL gradient locally at the behavior student. Arm A4 uses old and teacher trajectories in a two-source conditional flow-matching objective and exactly realizes their mixture of time marginals in the population limit.

The derivations stay at the level needed for flow-matching practice: every arm specifies its sampling distribution, target, loss, gradient, scale convention, exactness claim, and failure boundary. In particular, plain ODE MSE is not an unscaled zero-noise KL; it is a quadratic field/transport geometry, or equivalently a rescaled small-noise control-energy limit.

Contents

1 Common notation and reusable results

1.1 Flow, rollout, and scale conventions

We parameterize time so that sampling integrates from $t = 0$ (base noise) to $t = 1$ (generated sample). This is only a change of orientation from schedulers that denoise from 1 to 0. Let C denote conditioning information and $Z \sim \rho_{\text{base}}(\cdot \mid C)$ the initial noise. At outer iteration k, the frozen behavior student has vector field $v_{\text{old}} = v_{\theta_k}$ and rollout

$$\frac{\mathrm{d}X_t^{\text{old}}}{\mathrm{d}t} = v_{\text{old}}(t, X_t^{\text{old}}, C), \qquad X_0^{\text{old}} = Z. \tag{1}$$

The frozen teacher field is v_T, and v_θ is the trainable field. Their flow maps and time marginals are denoted by $\Phi_t^{\text{old}}, \Phi_t^T, \Phi_t^\theta$ and $\rho_{\text{old},t}, \rho_{T,t}, \rho_{\theta,t}$.

Let $\omega(t) > 0$ be a normalized training-time density, $\int_0^1 \omega(t)\mathrm{d}t = 1$. The old-occupancy expectation is

$$\mathbb{E}_{\text{old}}[f] := \mathbb{E}_{C,Z} \int_0^1 \omega(t) f(t, X_t^{\text{old}}, C)\mathrm{d}t. \tag{2}$$

The state X_t^{old} is detached during optimization. Thus (2) describes fixed-input field regression; it does not differentiate through the ODE solver.

For a detached positive-definite metric G_t, define

$$\|a\|_{G_t}^2 = a^\top G_t a, \qquad \|a\|_{\text{old},G}^2 = \mathbb{E}_{\text{old}}\left[\left\|a(t, X_t^{\text{old}}, C)\right\|_{G_t}^2\right]. \tag{3}$$

The default practical choice is $G_t = I$. A covariance-derived metric $G_t = A_t^{-1}$ is meaningful only when a reference covariance A_t has been explicitly declared; an ODE does not supply one automatically.

If the latent event has dimension D, then

$$\mathrm{RMS}_{G_t}(a)^2 := \frac{1}{D} \|a\|_{G_t}^2, \qquad \|a\|_{G_t}^2 = D \; \mathrm{RMS}_{G_t}(a)^2. \tag{4}$$

The event sum is the scale used by a joint Gaussian density. The RMS or event mean is easier to compare across resolutions. A trust radius or temperature must state which convention it uses; the two differ by D in squared scale.

1.2 Three different objects

The five arms act on three related but non-interchangeable objects:

object	notation	what is compared
one-step conditional kernel	$K(\mathrm{d}y \mid x, t)$	stochastic next-state law
vector field / flow map	$v(t, x)$, $\Phi_t(z)$	deterministic transport rule
time marginal	$\rho_t = (\Phi_t)_\# \rho_{\text{base}}$	distribution of states at time t

An SDE transition KL belongs to the first row. ODE velocity MSE belongs to the second. A marginal-mixture construction belongs to the third. Replacing one row by another can produce a useful surrogate, but it is not an algebraic identity.

1.3 Why deterministic transition KL is singular

For one numerical step of size h, let

$$F_j^h(x) = x + h\, v_j(t, x), \qquad K_j(\mathrm{d}y \mid x) = \delta_{F_j^h(x)}(\mathrm{d}y), \qquad j \in \{\text{old}, T\}. \tag{5}$$

Proposition 1.1 (No smooth conditional-KL trust region for unequal ODE maps)**.** *Assume* $F^h_{\text{old}}(x) \neq F^h_T(x)$ *and define*

$$Q_\alpha = (1-\alpha)K_{\text{old}} + \alpha K_T, \qquad \alpha \in (0,1).$$

Then

$$\mathrm{KL}(\delta_y \| Q_\alpha) = \begin{cases} -\log(1-\alpha), & y = F^h_{\text{old}}(x), \\ -\log\alpha, & y = F^h_T(x), \\ +\infty, & \textit{otherwise.} \end{cases} \tag{6}$$

Therefore the objective is finite at the old map but infinite after a generic arbitrarily small output perturbation. It has no local teacher-pull gradient.

Proof. Q_α places mass only at the two displayed atoms. The Radon–Nikodym derivative of δ_y with respect to Q_α is $1/(1-\alpha)$ at the old atom, $1/\alpha$ at the teacher atom, and does not exist elsewhere. Substitution into the definition of KL gives (6). □

JS divergence is finite here, but it is piecewise constant as the atoms move and supplies no useful local teacher-pull gradient. A deterministic PPO-style likelihood ratio is likewise undefined. The problem is the singular support, not merely the choice of divergence.

1.4 What does survive the small-noise limit

Smooth two outputs with a shared covariance εA, where $A \succ 0$:

$$p^\varepsilon_a = \mathcal{N}(a, \varepsilon A), \qquad p^\varepsilon_b = \mathcal{N}(b, \varepsilon A). \tag{7}$$

The equal-covariance Gaussian identity gives

$$\mathrm{KL}(p^\varepsilon_a \| p^\varepsilon_b) = \frac{1}{2\varepsilon}\,\|a-b\|^2_{A^{-1}}\,, \qquad 2\varepsilon\,\mathrm{KL}(p^\varepsilon_a \| p^\varepsilon_b) = \|a-b\|^2_{A^{-1}}\,. \tag{8}$$

The unscaled KL diverges as $\varepsilon \to 0$ for $a \neq b$, but the rescaled KL is exactly a quadratic transport cost.

For Euler means $\mu_j = x + h\,v_j$ and diffusion covariance $\Sigma = \varepsilon h A$,

$$\mathrm{KL}(\mathcal{N}(\mu_1, \varepsilon h A)\,\|\,\mathcal{N}(\mu_2, \varepsilon h A)) = \frac{h}{2\varepsilon}\,\|v_1 - v_2\|^2_{A^{-1}}\,. \tag{9}$$

Let P^{grid}_1 and P^{grid}_2 be the two Markov trajectory laws on the same grid and with the same initial law. The KL chain rule and (9) give

$$\begin{aligned} \varepsilon \mathrm{KL}(P^{\text{grid}}_1 \| P^{\text{grid}}_2) &= \frac{1}{2}\mathbb{E}_{P^{\text{grid}}_1} \sum_\ell h_\ell\,\|v_1(t_\ell, X_\ell) - v_2(t_\ell, X_\ell)\|^2_{A_\ell^{-1}} \\ &\longrightarrow \frac{1}{2}\mathbb{E}_{P_1} \int_0^1 \|v_1(t, X_t) - v_2(t, X_t)\|^2_{A_t^{-1}}\,\mathrm{d}t. \end{aligned} \tag{10}$$

Equation (10) is the principled bridge from SDE KL to ODE MSE. It preserves a rescaled pairwise control cost; it does *not* preserve the transition-mixture responsibility gate, as Section 5 will show.

1.5 A field-space trust region controls ODE displacement

Proposition 1.2 (Old-occupancy field bound)**.** *Assume $v_\theta(t,\cdot,c)$ is L-Lipschitz for every (t,c). Couple the candidate and old ODEs with the same (C,Z). Then*

$$\left\|X_1^\theta - X_1^{\mathrm{old}}\right\| \le \int_0^1 e^{L(1-s)} \left\|v_\theta(s, X_s^{\mathrm{old}}, C) - v_{\mathrm{old}}(s, X_s^{\mathrm{old}}, C)\right\| \mathrm{d}s, \tag{11}$$

$$\mathbb{E}\left\|X_1^\theta - X_1^{\mathrm{old}}\right\|^2 \le e^{2L} \int_0^1 \mathbb{E}\left\|v_\theta(s, X_s^{\mathrm{old}}, C) - v_{\mathrm{old}}(s, X_s^{\mathrm{old}}, C)\right\|^2 \mathrm{d}s. \tag{12}$$

Consequently,

$$W_2^2(\rho_{\theta,1}, \rho_{\mathrm{old},1}) \le \mathbb{E}\left\|X_1^\theta - X_1^{\mathrm{old}}\right\|^2. \tag{13}$$

If additionally

$$\omega(t) \ge \omega_* > 0, \qquad G_t \succeq g_* I \tag{14}$$

for all trained times, then the field norm in (3) *controls the endpoint:*

$$W_2^2(\rho_{\theta,1}, \rho_{\mathrm{old},1}) \le \frac{e^{2L}}{\omega_* g_*} \left\|v_\theta - v_{\mathrm{old}}\right\|_{\mathrm{old},G}^2. \tag{15}$$

Proof. Add and subtract $v_\theta(t, X_t^{\mathrm{old}}, C)$ in the difference of the two ODEs. The candidate-field Lipschitz bound and Grönwall's inequality give (11). Cauchy–Schwarz gives (12). The common-noise pairing is a valid coupling of the two endpoint laws, proving (13). Finally, *under the additional lower bounds* in (14),

$$\left\|v_\theta - v_{\mathrm{old}}\right\|_{\mathrm{old},G}^2 \ge \omega_* g_* \int_0^1 \mathbb{E}\left\|v_\theta(t, X_t^{\mathrm{old}}) - v_{\mathrm{old}}(t, X_t^{\mathrm{old}})\right\|^2 \mathrm{d}t,$$

which gives (15). □

Remark 1.3. The bound controls departure from the *old* rollout because the field error is evaluated at old states. Replacing it by a candidate–teacher endpoint guarantee requires candidate-occupancy teacher error, or an additional coverage/Lipschitz argument. Old-state MSE alone does not give a global teacher-distribution convergence theorem.

2 Arm A0: Direct ODE-OPD

2.1 Objective and gradient

Direct ODE-OPD queries the teacher on states visited by the behavior student and regresses the current field to the teacher:

$$\mathcal{L}_{\mathrm{A0}}(\theta) = \frac{1}{2}\left\|v_\theta - v_T\right\|_{\mathrm{old},G}^2 = \frac{1}{2}\mathbb{E}_{\mathrm{old}}\left[\left\|v_\theta(t, X_t^{\mathrm{old}}, C) - v_T(t, X_t^{\mathrm{old}}, C)\right\|_{G_t}^2\right]. \tag{16}$$

This is the same direct-matching family as DiffusionOPD/XOPD [2]: rollout is on-policy, while optimization is closed-form regression and does not require transition likelihoods. Their deterministic transition-mean objective corresponds specifically to the h_ℓ^2-weighted velocity loss derived below, rather than to an arbitrary choice of (ω, G) in (16).

Proposition 2.1 (A0 is old-occupancy weighted field regression)**.** *In unrestricted output space, every attained population minimizer of* (16) *satisfies*

$$v^\star(t,x,c) = v_T(t,x,c) \tag{17}$$

almost everywhere under the old occupation measure. In a model class $\mathcal{V}$, any attained global solution is the weighted least-squares fit

$$v^\star_{\mathcal{V}} \in \arg\min_{v\in\mathcal{V}} \|v - v_T\|^2_{\mathrm{old},G}\,. \tag{18}$$

Its parameter gradient is

$$\nabla_\theta \mathcal{L}_{\mathrm{A0}} = \mathbb{E}_{\mathrm{old}}\Big[J_\theta^\top G_t(v_\theta - v_T)\Big]\,, \qquad J_\theta = \partial_\theta v_\theta. \tag{19}$$

Proof. Positive definiteness gives $\|v - v_T\|^2_{G_t} \geq 0$, with equality exactly at $v = v_T$. Integration preserves this characterization almost everywhere under the occupation measure. Restricting the minimization to $\mathcal{V}$ gives (18); differentiating the detached-input quadratic gives (19). □

For a linear output model $v_\theta = B(t,x,c)\theta$, the finite-capacity optimum obeys the weighted normal equations

$$\mathbb{E}_{\mathrm{old}}[B^\top G_t B]\,\theta^\star = \mathbb{E}_{\mathrm{old}}[B^\top G_t v_T]. \tag{20}$$

For a nonlinear network, stationarity only implies tangent-space orthogonality

$$\mathbb{E}_{\mathrm{old}}\Big[J_{\theta^\star}^\top G_t(v_{\theta^\star} - v_T)\Big] = 0; \tag{21}$$

it does not imply existence or uniqueness of a global least-squares solution.

2.2 The practical loss space

For an Euler step and a linear flow-matching path,

$$\mu_\theta - \mu_T = h(v_\theta - v_T), \qquad \widehat{x}_0^\theta - \widehat{x}_0^T = s(v_\theta - v_T). \tag{22}$$

The plus sign follows from the common noise-to-data time orientation: $x_s = x_0 - sv$ and hence $\widehat{x}_0 = x_s + sv$. A native data-to-noise predictor $u = \varepsilon - x_0 = -v$ uses the opposite sign; the squared-loss scale below is unchanged. Therefore velocity, next-state mean, and clean-prediction MSE differ only by timestep weights:

$$\mathcal{L}_v : \mathcal{L}_{x_t} : \mathcal{L}_{x_0} \quad \Longleftrightarrow \quad 1 : h^2 : s^2. \tag{23}$$

At one fixed timestep these factors only rescale the gradient. Across a trajectory they define different quadrature rules. In particular,

$$\int_0^1 \|\Delta v_t\|^2\,\mathrm{d}t \approx \sum_\ell h_\ell\,\|\Delta v_\ell\|^2 = \sum_\ell \frac{\|\Delta\mu_\ell\|^2}{h_\ell}, \tag{24}$$

whereas an unnormalized sum of next-state MSE uses h_ℓ^2 and tends to zero under grid refinement.

2.3 What A0 does and does not guarantee

If the fitted field remains close to the old field, Proposition 1.2 turns measured field displacement into a bound on rollout displacement. A0 itself, however, contains no old-field penalty. The triangle inequality only gives

$$\|v_\theta - v_{\mathrm{old}}\|_{\mathrm{old},G} \leq \|v_\theta - v_T\|_{\mathrm{old},G} + \|v_T - v_{\mathrm{old}}\|_{\mathrm{old},G}\,, \tag{25}$$

which can be large when the teacher is far from the student.

The loss also constrains only old-occupancy regions. Low-density or teacher-only states receive little or no supervision in a finite batch. Refreshing the student rollout after every outer iteration improves coverage adaptively, but does not turn a local regression result into a global distribution contraction.

Exact interpretation of Arm A0

A0 is on-policy *field regression*: the inputs come from the current behavior flow. It is not proximal merely because those inputs are on-policy, and it is not an ODE transition KL because unequal deterministic kernels are singular by Proposition 1.1.

3 Arm A1: Transport-Projected ODE-OPD

3.1 Quadratic proximal target

Arm A1 explicitly anchors the field to the behavior student:

$$\mathcal{L}_{\mathrm{A1},\alpha}(v) = \frac{\alpha}{2} \left\| v - v_T \right\|_{\mathrm{old},G}^2 + \frac{1-\alpha}{2} \left\| v - v_{\mathrm{old}} \right\|_{\mathrm{old},G}^2, \qquad \alpha \in [0,1]. \tag{26}$$

Proposition 3.1 (Barycentric target)**.** *Define*

$$v_\alpha = (1-\alpha)v_{\mathrm{old}} + \alpha v_T, \qquad D_k = \left\| v_T - v_{\mathrm{old}} \right\|_{\mathrm{old},G}. \tag{27}$$

Then

$$\mathcal{L}_{\mathrm{A1},\alpha}(v) = \frac{1}{2} \left\| v - v_\alpha \right\|_{\mathrm{old},G}^2 + \frac{\alpha(1-\alpha)}{2} D_k^2. \tag{28}$$

Thus v_α is the unrestricted optimum. In a finite neural class, training is a weighted least-squares fit to v_α, not a parameter interpolation between the two networks.

Proof. Apply pointwise

$$\alpha \left\| v - a \right\|_G^2 + (1-\alpha) \left\| v - b \right\|_G^2 = \left\| v - \alpha a - (1-\alpha) b \right\|_G^2 + \alpha(1-\alpha) \left\| a - b \right\|_G^2$$

and average under the old occupancy. □

3.2 Equivalent trust-budget projection

Consider the explicit field-space trust problem

$$\min_v \ \frac{1}{2} \left\| v - v_T \right\|_{\mathrm{old},G}^2 \quad \text{subject to} \quad \frac{1}{2} \left\| v - v_{\mathrm{old}} \right\|_{\mathrm{old},G}^2 \le \delta. \tag{29}$$

Proposition 3.2 (Closed-form trust projection)**.** *Let $R = \sqrt{2\delta}$. If $D_k > 0$, the unrestricted solution of* (29) *is*

$$v_R = v_{\mathrm{old}} + \alpha_R (v_T - v_{\mathrm{old}}), \qquad \alpha_R = \min\left\{1, \frac{R}{D_k}\right\}. \tag{30}$$

If $D_k = 0$, $v_R = v_{\mathrm{old}} = v_T$. Conversely, a fixed α corresponds to

$$R_\alpha = \alpha D_k, \qquad \delta_\alpha = \frac{\alpha^2 D_k^2}{2}. \tag{31}$$

Proof. Write $u = v - v_{\mathrm{old}}$ and $d = v_T - v_{\mathrm{old}}$. The objective is

$$\left\| u - d \right\|_{\mathrm{old},G}^2 = \left\| u \right\|_{\mathrm{old},G}^2 + D_k^2 - 2 \left\langle u, d \right\rangle_{\mathrm{old},G}.$$

For fixed $\left\| u \right\|_{\mathrm{old},G}$, Cauchy–Schwarz makes u parallel to d optimal. The best allowed length is $\min\{R, D_k\}$, giving (30). □

A fixed α is therefore a *relative* trust radius: the absolute update grows with the teacher–old gap. If a fixed absolute field RMS radius r is desired, use

$$
\begin{aligned}
d_k^{\mathrm{RMS}} &= \sqrt{\mathbb{E}_{\mathrm{old}}[\mathrm{RMS}_{G_t}(v_T - v_{\mathrm{old}})^2]}, \\
\alpha_k &= \begin{cases} 1, & d_k^{\mathrm{RMS}} = 0, \\ \min\{1, r/d_k^{\mathrm{RMS}}\}, & d_k^{\mathrm{RMS}} > 0. \end{cases}
\end{aligned} \tag{32}
$$

3.3 Wasserstein projection of a stochastic branch target

At one old state, define the old and teacher Euler outputs

$$y_{\mathrm{old}} = x + hv_{\mathrm{old}}(t,x), \qquad y_T = x + hv_T(t,x), \tag{33}$$

and the branch mixture

$$Q_{\alpha,x} = (1-\alpha)\delta_{y_{\mathrm{old}}} + \alpha\delta_{y_T}. \tag{34}$$

Proposition 3.3 (Best deterministic quadratic-transport projection)**.** *For the quadratic ground metric G_t,*

$$\arg\min_y W_{2,G_t}^2(\delta_y, Q_{\alpha,x}) = (1-\alpha)y_{\mathrm{old}} + \alpha y_T, \tag{35}$$

$$\min_y W_{2,G_t}^2(\delta_y, Q_{\alpha,x}) = \alpha(1-\alpha)\left\|y_T - y_{\mathrm{old}}\right\|_{G_t}^2. \tag{36}$$

Proof. The only coupling from δ_y to $Q_{\alpha,x}$ sends masses $1-\alpha$ and α to the two atoms, so

$$W_{2,G_t}^2(\delta_y, Q_{\alpha,x}) = (1-\alpha)\left\|y - y_{\mathrm{old}}\right\|_{G_t}^2 + \alpha\left\|y - y_T\right\|_{G_t}^2.$$

Completing the square yields (35) and (36). □

For Euler maps, the minimizer is $x + hv_\alpha$ and the irreducible cost is

$$\alpha(1-\alpha)h^2\left\|v_T - v_{\mathrm{old}}\right\|_{G_t}^2. \tag{37}$$

Thus A1 is the best deterministic approximation to a local stochastic branch mixture under the chosen quadratic metric. It does *not* reproduce the two-atom law.

3.4 Why one update is only a scaled A0 update

Proposition 3.4 (First-step anchor degeneracy)**.** *Assume exact replay, so $v_{\theta_k} = v_{\mathrm{old}}$ on every stored state. Then*

$$\nabla_\theta \mathcal{L}_{\mathrm{A1},\alpha}\big|_{\theta_k} = \alpha\, \nabla_\theta \mathcal{L}_{\mathrm{A0}}\big|_{\theta_k}. \tag{38}$$

Proof. The old-anchor gradient contains $J_{\theta_k}^\top G_t(v_{\theta_k} - v_{\mathrm{old}})$, which is pointwise zero. Only the teacher term remains, multiplied by α. □

For plain SGD, one step is therefore A0 with learning rate $\alpha\eta$. Adam moments, clipping, weight decay, and the optimizer epsilon can prevent the final parameter update from scaling exactly by α, but the loss gradient identity remains exact. If output-space gradient descent with $G_t = I$ and step size $0 < \eta < 2$ is repeated against a frozen old target, then

$$v^{(m)} = v_{\mathrm{old}} + \alpha\big[1 - (1-\eta)^m\big](v_T - v_{\mathrm{old}}). \tag{39}$$

The old anchor becomes active after the first update and the limit is v_α. Genuine first-update trust enforcement instead requires explicit projection, a constrained line search, or post-update stopping based on the measured field displacement.

Practical meaning of Arm A1

A1 supplies a real geometric trust region only when the radius is enforced or the anchored subproblem is optimized beyond its first gradient. With one on-policy optimizer step, a fixed α is only a teacher-gradient scale. Flow-DPPO [4] motivates controlling current–old divergence; A1 is a field-space analogue with a local one-step transport interpretation, not a deterministic likelihood ratio or a full-flow-map projection.

4 Arm A2: Forward-Risk P-OPD

4.1 Forward-process evidence

Arm A2 keeps efficient ODE rollout but constructs a supervised flow-matching test after the rollout. Let $x_{\mathrm{data}} = X_1^{\mathrm{old}}$ be the old ODE endpoint, draw $s \sim \omega_{\mathrm{FM}}$ on $[0,1]$ and $\varepsilon \sim \mathcal{N}(0, I_D)$, and form

$$x_s = (1-s)x_{\mathrm{data}} + s\varepsilon = x_{\mathrm{data}} + su, \qquad u = \varepsilon - x_{\mathrm{data}}. \tag{40}$$

Let $z = (x_s, s, C)$ and evaluate frozen predictions $f_{\mathrm{old}}(z)$ and $f_T(z)$ of the native data-to-noise target u on the same input. Every A2 expectation below is over (C, Z, s, ε) under this construction. This is the solver-decoupled forward-flow-matching construction [1] shared by DPO, DiffusionNFT, AWM, DGPO, and CRD.

To make event scaling explicit, define

$$E_j(z,u) = \frac{1}{2c_D}\left\|u - f_j(z)\right\|^2, \qquad c_D = \begin{cases} 1, & \text{event sum,} \\ D, & \text{event mean,} \end{cases} \tag{41}$$

and

$$A_T = E_{\mathrm{old}} - E_T, \qquad \gamma_T = \sigma\left(\operatorname{logit}\alpha + \frac{A_T}{\tau_s}\right). \tag{42}$$

Positive A_T means that the teacher predicts the sampled forward target better than the old student.

4.2 Exact auxiliary Gaussian interpretation

Proposition 4.1 (When the forward-risk gate is a Bayes responsibility)**.** *Fix z and introduce a component label $B \in \{\mathrm{old}, T\}$ with $\Pr(B = T) = \alpha$. Declare the predictive likelihood*

$$p_j(u \mid z) = (2\pi\sigma_r^2(s))^{-D/2} \exp\left[-\frac{\|u - f_j(z)\|^2}{2\sigma_r^2(s)}\right]. \tag{43}$$

Then

$$\Pr(B = T \mid u, z) = \sigma\left(\operatorname{logit}\alpha + \frac{c_D A_T}{\sigma_r^2(s)}\right). \tag{44}$$

Therefore (42) *is exactly this posterior when*

$$\tau_s = \frac{\sigma_r^2(s)}{c_D}. \tag{45}$$

Proof. The common Gaussian normalizers cancel and

$$\log\frac{p_T(u \mid z)}{p_{\mathrm{old}}(u \mid z)} = \frac{\|u - f_{\mathrm{old}}\|^2 - \|u - f_T\|^2}{2\sigma_r^2(s)} = \frac{c_D A_T}{\sigma_r^2(s)}.$$

Bayes' rule for the two-component mixture gives (44). □

The covariance $\sigma_r^2(s)I$ in Proposition 4.1 is a *declared predictive-residual model*; it is not automatically equal to the variance used to create x_s. Unequal component covariances would add log-determinant and quadratic terms that a plain square-loss difference does not contain. Using an event mean with an unchanged numerical temperature is equivalent to inflating the predictive covariance by D.

4.3 What risk difference estimates

Proposition 4.2 (Conditional square-loss decomposition)**.** *Let*

$$m_{\text{old}}(z) = \mathbb{E}[u \mid z], \qquad V_{\text{old}}(z) = \mathbb{E}\left[\|u - m_{\text{old}}(z)\|^2 \mid z\right], \tag{46}$$

where the expectation is under old-generated endpoints and fresh forward noise. Then

$$\mathbb{E}[E_j \mid z] = \frac{1}{2c_D}\left(V_{\text{old}}(z) + \|f_j(z) - m_{\text{old}}(z)\|^2\right) \tag{47}$$

and

$$\mathbb{E}[A_T \mid z] = \frac{\|f_{\text{old}} - m_{\text{old}}\|^2 - \|f_T - m_{\text{old}}\|^2}{2c_D}. \tag{48}$$

Proof. Write $u - f_j = (u - m_{\text{old}}) + (m_{\text{old}} - f_j)$. The conditional cross term vanishes because $\mathbb{E}[u - m_{\text{old}} \mid z] = 0$. Subtracting the two decompositions cancels the same irreducible conditional variance. □

Thus A_T is a one-sample unbiased estimate of the conditional *risk difference*. The nonlinear gate is not an unbiased estimate of the sigmoid of the mean risk difference:

$$\mathbb{E}[\gamma_T \mid z] \neq \sigma\left(\operatorname{logit}\alpha + \frac{\mathbb{E}[A_T \mid z]}{\tau_s}\right) \quad \text{in general.} \tag{49}$$

Also, the old ODE field need not equal $m_{\text{old}}(z)$ after an endpoint has been re-noised with fresh ε. The construction compares two predictors on old-generated data; it does not assume that the old predictor is Bayes-optimal for the new coupling.

4.4 Gated projected target and trust radius

Let $r_s > 0$ be an event-L^2 output radius in metric G_s. A per-dimension RMS radius r_s^{RMS} corresponds to $r_s = \sqrt{D}\, r_s^{\text{RMS}}$. Define

$$d(z) = f_T(z) - f_{\text{old}}(z), \qquad c(z) = \begin{cases} 1, & \|d(z)\|_{G_s} = 0, \\ \min\{1, r_s / \|d(z)\|_{G_s}\}, & \|d(z)\|_{G_s} > 0. \end{cases} \tag{50}$$

The detached target and loss are

$$f_\gamma(z,u) = f_{\text{old}}(z) + \operatorname{sg}(\gamma_T)c(z)d(z), \tag{51}$$

$$\mathcal{L}_{\text{A2}}(\theta) = \mathbb{E}\left[\frac{1}{2c_D}\|f_\theta(z) - f_\gamma(z,u)\|_{G_s}^2\right]. \tag{52}$$

Since $0 \le \gamma_T, c \le 1$,

$$\|f_\gamma - f_{\text{old}}\|_{G_s} \le r_s. \tag{53}$$

At exact replay, $f_{\theta_{\text{old}}} = f_{\text{old}}$, and

$$\nabla_\theta \mathcal{L}_{\text{A2}}|_{\theta_{\text{old}}} = -\frac{1}{c_D}\mathbb{E}\left[\gamma_T c(z) J_{\text{old}}^\top G_s (f_T - f_{\text{old}})\right]. \tag{54}$$

The gate decides whether the teacher is useful on this supervised forward target; $c(z)$ decides how far the target is allowed to move.

For any $\delta \in (0, \frac{1}{2})$,

$$\gamma_T \le \delta \iff A_T \le \tau_s\big(\operatorname{logit}\delta - \operatorname{logit}\alpha\big), \tag{55}$$
$$\gamma_T \ge 1-\delta \iff A_T \ge \tau_s\big(\operatorname{logit}(1-\delta) - \operatorname{logit}\alpha\big). \tag{56}$$

An event sum normally scales as D at a fixed per-coordinate risk gap, so a fixed τ_s becomes increasingly binary with resolution. An event mean is stable across D but changes the auxiliary Gaussian model. A practical implementation should log both A_T and A_T/D, gate entropy and saturation, $\|d\|_{\mathrm{RMS}}$, $\|d\|_2$, $c(z)$, and the realized post-update field displacement.

Exact boundary of Arm A2

The Gaussian statement concerns an auxiliary likelihood for the target u conditioned on z. Neither $\rho_T(x_s)$ nor $\rho_{\mathrm{old}}(x_s)$ appears. Therefore the gate is predictive risk evidence on old-generated samples, not a teacher/old state-density ratio and not an exact marginal-mixture objective. Timestep centering, running-scale normalization, or a calibrated temperature can be useful, but they define an explicit surrogate.

5 Arm A3: SDE Transition-Mixture P-OPD

5.1 Exact one-step mixture

Unlike Sections 2–4, Arm A3 uses a frozen behavior *SDE* rollout. On a grid, sample

$$X^{\mathrm{old}}_{\ell+1} \sim p_{\mathrm{old},\ell}(\cdot \mid X^{\mathrm{old}}_\ell, C), \qquad J \sim \nu_{\mathrm{train}}, \tag{57}$$

and let $\mathbb{E}_{\mathrm{SDE-old}}$ denote expectation over $(C, X^{\mathrm{old}}_{0:L}, J)$. The next state $Y = X^{\mathrm{old}}_{J+1}$ must be the actual transition sampled in this rollout.

Condition on one $(C, J, X^{\mathrm{old}}_J)$. The old, teacher, and current one-step kernels share a positive, model-independent covariance:

$$p_j(y) = \mathcal{N}(y \mid \mu_j, \Sigma), \qquad j \in \{\mathrm{old}, T, \theta\}, \qquad \Sigma \succ 0. \tag{58}$$

Define

$$q_\alpha(y) = (1-\alpha)p_{\mathrm{old}}(y) + \alpha p_T(y), \qquad Y \sim p_{\mathrm{old}}, \tag{59}$$

and

$$\gamma(Y) = \frac{\alpha p_T(Y)}{q_\alpha(Y)} = \sigma\left(\operatorname{logit}\alpha + \log\frac{p_T(Y)}{p_{\mathrm{old}}(Y)}\right). \tag{60}$$

This is the flow-transition version of TOP-D's external probability-mixture teacher [3].

Proposition 5.1 (Exact local mixture-KL gradient)**.** *Let $\mathcal{K}(\mu_\theta) = \mathrm{KL}(p_\theta \| q_\alpha)$. At the behavior mean,*

$$\nabla_{\mu_\theta}\mathcal{K}\big|_{\mu_\theta=\mu_{\mathrm{old}}} = \mathbb{E}_{p_{\mathrm{old}}}[\gamma(Y)]\Sigma^{-1}(\mu_{\mathrm{old}} - \mu_T). \tag{61}$$

Proof. Write $Y = \mu_\theta + \Sigma^{1/2}Z$ with $Z \sim \mathcal{N}(0, I)$. Under this reparameterization, $\log p_\theta(Y)$ is independent of μ_θ, so

$$\nabla_{\mu_\theta}\mathcal{K} = -\mathbb{E}_{p_\theta}[\nabla_y \log q_\alpha(Y)].$$

The mixture score is

$$\nabla_y \log q_\alpha(y) = \Sigma^{-1}((1-\gamma(y))\mu_{\mathrm{old}} + \gamma(y)\mu_T - y)\,.$$

At $\mu_\theta = \mu_{\mathrm{old}}$, the expectation of $Y - \mu_{\mathrm{old}}$ vanishes, leaving (61). □

Now define the detached local surrogate

$$\mathcal{S}_{\text{sum}}(\mu_\theta) = \mathbb{E}_{Y\sim p_{\text{old}}}\left[\operatorname{sg}(\gamma(Y))\frac{1}{2}\left\|\mu_\theta - \mu_T\right\|^2_{\Sigma^{-1}}\right]. \tag{62}$$

Its gradient at $\mu_\theta = \mu_{\text{old}}$ is exactly (61). If μ_θ is a network output, both gradients are left-multiplied by the same Jacobian $\partial_\theta \mu_\theta^\top$.

The complete Arm A3 event-sum loss averages this conditional surrogate over the actual behavior SDE trajectory:

$$\mathcal{L}_{\text{A3,sum}}(\theta) = \mathbb{E}_{\text{SDE}-\text{old}}\left[\operatorname{sg}(\gamma_J(Y))\frac{1}{2}\left\|\mu_{\theta,J}(X_J^{\text{old}}, C) - \mu_{T,J}(X_J^{\text{old}}, C)\right\|^2_{\Sigma_J^{-1}}\right]. \tag{63}$$

The fixed-state theorem applies inside this outer expectation. Replacing $\mathbb{E}_{\text{SDE}-\text{old}}$ by the ODE occupation measure (2) would remove the sampled stochastic transition and invalidate the density-ratio construction.

An event-mean implementation uses

$$\mathcal{S}_{\text{mean}} = \frac{1}{D}\mathcal{S}_{\text{sum}}, \qquad \nabla\mathcal{S}_{\text{mean}} = \frac{1}{D}\nabla\mathcal{K} \quad \text{at } \mu_\theta = \mu_{\text{old}}. \tag{64}$$

It matches the gradient of the dimension-normalized objective $\mathcal{K}/D$, not the numerical gradient of the unnormalized joint KL.

5.2 Ratio law and overlap

Proposition 5.2 (Gaussian ratio law)**.** *Fix $\alpha \in (0,1)$ and let*

$$\Delta = \mu_T - \mu_{\text{old}}, \qquad K = \frac{1}{2}\Delta^\top\Sigma^{-1}\Delta = \mathrm{KL}(p_{\text{old}}\|p_T). \tag{65}$$

Under $Y \sim p_{\text{old}}$,

$$\log\frac{p_T(Y)}{p_{\text{old}}(Y)} \sim \mathcal{N}(-K, 2K). \tag{66}$$

Moreover,

$$\mathbb{E}_{p_{\text{old}}}[\gamma(Y)] \le \min\left\{\alpha,\ \frac{1}{2}\sqrt{\frac{\alpha}{1-\alpha}}e^{-K/4}\right\}. \tag{67}$$

Proof. Write $Y = \mu_{\text{old}} + \Sigma^{1/2}Z$. Direct expansion gives

$$\log\frac{p_T(Y)}{p_{\text{old}}(Y)} = Z^\top\Sigma^{-1/2}\Delta - K.$$

The first term has variance $\Delta^\top\Sigma^{-1}\Delta = 2K$. For the second claim, put $r = p_T/p_{\text{old}}$. The arithmetic–geometric mean inequality gives

$$\gamma = \frac{\alpha r}{1-\alpha+\alpha r} \le \frac{1}{2}\sqrt{\frac{\alpha}{1-\alpha}}\sqrt{r}.$$

Integrating under p_{old} gives the Gaussian Bhattacharyya coefficient

$$\int\sqrt{p_{\text{old}}p_T} = \exp\left(-\frac{1}{8}\Delta^\top\Sigma^{-1}\Delta\right) = e^{-K/4}.$$

Finally, concavity of $r \mapsto \alpha r/(1-\alpha+\alpha r)$ and $\mathbb{E}_{p_{\text{old}}}[r] = 1$ give the bound by α. $\square$

Equation (67) makes the mechanism explicit: the expected teacher pull is controlled by overlap between the two complete transition events, not merely by per-coordinate agreement.

5.3 The ODE limit: the gate disappears, the control cost survives

Keep $\alpha \in (0,1)$ fixed and let

$$\mu_j = x + hF_j, \qquad \Sigma_\varepsilon = \varepsilon h A, \qquad A \succ 0. \tag{68}$$

Then

$$K_\varepsilon = \frac{h}{2\varepsilon} \left\| F_T - F_{\text{old}} \right\|_{A^{-1}}^2 . \tag{69}$$

If $F_T \neq F_{\text{old}}$ and $h/\varepsilon \to \infty$, then

$$\gamma(Y) \to 0 \quad \text{in probability}, \qquad \mathbb{E}[\gamma(Y)] = O\left(\exp\left[-\frac{h}{8\varepsilon} \left\| F_T - F_{\text{old}} \right\|_{A^{-1}}^2\right]\right). \tag{70}$$

The mean-gradient with respect to the drift is

$$\nabla_{F_\theta} \mathcal{K}\big|_{F_\theta = F_{\text{old}}} = \frac{h}{\varepsilon} \mathbb{E}[\gamma(Y)] A^{-1}(F_{\text{old}} - F_T), \tag{71}$$

so the exponential overlap loss dominates the polynomial $1/\varepsilon$ factor and the teacher gradient vanishes.

This limit requires $K_\varepsilon \to \infty$. Taking $h \to 0$ first at fixed nonzero diffusion intensity instead makes the one-step K_ε small. The relevant ODE statement is the zero-noise limit at a fixed practical solver grid, or any joint limit with $h/\varepsilon \to \infty$.

In contrast, the pairwise KL obeys

$$\frac{\varepsilon}{h} \text{KL}(\mathcal{N}(x + hF_\theta, \varepsilon h A) \,\|\, \mathcal{N}(x + hF_T, \varepsilon h A)) = \frac{1}{2} \left\| F_\theta - F_T \right\|_{A^{-1}}^2 , \tag{72}$$

which is the control-energy geometry of (10). The transition-mixture KL itself instead approaches a flat component-selection cost:

$$\text{KL}(p_{\text{old}} \| q_\alpha) \longrightarrow -\log(1-\alpha). \tag{73}$$

To verify the limit, put $r = p_T / p_{\text{old}}$ and write

$$\text{KL}(p_{\text{old}} \| q_\alpha) = -\log(1-\alpha) - \mathbb{E}_{p_{\text{old}}} \log\left(1 + \frac{\alpha}{1-\alpha} r\right). \tag{74}$$

Using $\log(1+x) \le 2\sqrt{x}$ and the Bhattacharyya coefficient from Proposition 5.2,

$$0 \le -\log(1-\alpha) - \text{KL}(p_{\text{old}} \| q_\alpha) \le 2\sqrt{\frac{\alpha}{1-\alpha}} e^{-K_\varepsilon/4}, \tag{75}$$

which tends to zero when $K_\varepsilon \to \infty$. Thus the quadratic metric has a useful deterministic limit, while the probability-mixture gate does not.

5.4 Temperature, latent sum, and latent mean

For equal-covariance Gaussians and $T_g > 0$,

$$\frac{1}{T_g} \log \frac{\mathcal{N}(y \mid \mu_T, \Sigma)}{\mathcal{N}(y \mid \mu_{\text{old}}, \Sigma)} = \log \frac{\mathcal{N}(y \mid \mu_T, T_g \Sigma)}{\mathcal{N}(y \mid \mu_{\text{old}}, T_g \Sigma)}. \tag{76}$$

Therefore dividing the log ratio by T_g is pointwise equivalent to inflating both component covariances by T_g. A fully consistent inflated-covariance mixture would also sample $Y \sim \mathcal{N}(\mu_{\text{old}}, T_g \Sigma)$ and divide the local quadratic by T_g. Changing only the gate temperature while retaining original samples and loss normalization is a tempered surrogate, not the exact gradient of the original mixture KL.

Define the whitened per-coordinate gap

$$w^2 = \frac{1}{D}\Delta^\top \Sigma^{-1}\Delta. \tag{77}$$

Then

$$K = \frac{D}{2}w^2, \qquad L_{\text{sum}} \sim \mathcal{N}\left(-\frac{D}{2}w^2, Dw^2\right), \tag{78}$$

while

$$L_{\text{mean}} = \frac{L_{\text{sum}}}{D} \sim \mathcal{N}\left(-\frac{1}{2}w^2, \frac{w^2}{D}\right). \tag{79}$$

The sum is the exact joint event likelihood. The mean is exactly gate temperature $T_g = D$, or covariance inflation by D. As D grows at fixed w, the sum gate separates exponentially; the mean gate converges to

$$\sigma\left(\operatorname{logit}\alpha - \frac{1}{2}w^2\right) \tag{80}$$

and becomes deterministic across samples. Intermediate temperatures define surrogates with an explicit effective dimension. This reproduces the companion exact-sum gate analysis.

Exact boundary of Arm A3

The local-gradient equality requires: the same complete latent event, positive shared model-independent covariance, an actual transition sampled by the frozen old student, $\mu_\theta = \mu_{\text{old}}$ at the expansion point, detached old/teacher/gate quantities, and the untempered joint ratio. It is an equality of first derivatives at one on-policy point, not an equality of loss values, Hessians, multi-step trajectory objectives, or gradients after multiple stale-data updates.

6 Arm A4: Marginal-Mixture Conditional Flow Matching

6.1 Mixture marginals and their velocity

Fix a conditioning value $C = c$. All densities and fields in this section are conditional on c, which is suppressed from the notation; the final training loss also averages over C. Assume the old and teacher ODE marginals have densities satisfying their continuity equations:

$$\partial_t \rho_{b,t} + \nabla_x \cdot (\rho_{b,t} v_b) = 0, \qquad \rho_{b,0} = \rho_{\text{base}}, \qquad b \in \{\text{old}, T\}. \tag{81}$$

Define

$$q_t = (1-\alpha)\rho_{\text{old},t} + \alpha\rho_{T,t}, \tag{82}$$

$$q_t u_t = (1-\alpha)\rho_{\text{old},t} v_{\text{old}} + \alpha\rho_{T,t} v_T. \tag{83}$$

Proposition 6.1 (Exact mixture continuity equation)**.** *The pair (q_t, u_t) satisfies*

$$\partial_t q_t + \nabla_x \cdot (q_t u_t) = 0, \qquad q_0 = \rho_{\text{base}}. \tag{84}$$

Where $q_t(x) > 0$, define

$$\gamma_t(x) = \frac{\alpha\rho_{T,t}(x)}{q_t(x)}. \tag{85}$$

Then

$$u_t(x) = (1-\gamma_t(x))v_{\text{old}}(t,x) + \gamma_t(x)v_T(t,x). \tag{86}$$

Proof. Substitute (82)–(83):

$$\partial_t q_t + \nabla\cdot(q_t u_t) = (1-\alpha)[\partial_t \rho_{\mathrm{old},t} + \nabla\cdot(\rho_{\mathrm{old},t} v_{\mathrm{old}})] \\ + \alpha[\partial_t \rho_{T,t} + \nabla\cdot(\rho_{T,t} v_T)] = 0.$$

The common base law gives $q_0 = \rho_{\mathrm{base}}$. Dividing the flux by q_t gives (86). □

This is the deterministic analogue of a probability-mixture responsibility, but the responsibility is over time marginals rather than one-step Dirac kernels.

6.2 Two-source CFM without density ratios

Sample one source branch for a whole trajectory and a training time:

$$B \sim \mathrm{Bernoulli}(\alpha), \qquad t \sim \omega, \qquad X_t = \Phi_t^B(Z), \qquad V_t = \begin{cases} v_{\mathrm{old}}(t, X_t), & B = 0, \\ v_T(t, X_t), & B = 1. \end{cases} \tag{87}$$

The practical objective is

$$\mathcal{L}_{\mathrm{A4}}(\theta) = \mathbb{E}_{C,t,B,Z} \left\| v_\theta(t, X_t) - V_t \right\|^2. \tag{88}$$

The norm in (88) is an event sum. A per-dimension implementation divides the entire loss, including the constant below, by D.

Proposition 6.2 (Two-source CFM identity)**.** *Under* (87)*,* $X_t \sim q_t$*,*

$$\Pr(B = 1 \mid X_t = x, t) = \gamma_t(x), \qquad \mathbb{E}[V_t \mid X_t = x, t] = u_t(x), \tag{89}$$

and

$$\mathcal{L}_{\mathrm{A4}}(\theta) = \mathbb{E}_{C,t,X_t \sim q_t(\cdot|C)} \left\| v_\theta(t, X_t) - u_t(X_t) \right\|^2 + \mathcal{C}_{\mathrm{var}}, \tag{90}$$

where

$$\mathcal{C}_{\mathrm{var}} = \mathbb{E}_{C,t,X_t \sim q_t(\cdot|C)} \left[\gamma_t(X_t)(1-\gamma_t(X_t)) \left\| v_T(t, X_t) - v_{\mathrm{old}}(t, X_t) \right\|^2 \right] \tag{91}$$

is independent of θ*.*

Proof. Marginalizing the branch label gives $X_t \sim (1-\alpha)\rho_{\mathrm{old},t} + \alpha\rho_{T,t} = q_t$. Bayes' rule gives (85), and the conditional mean of the two possible velocity labels is (86). Finally write

$$v_\theta - V_t = (v_\theta - u_t) + (u_t - V_t).$$

Conditioned on (t, X_t), the cross term vanishes because $\mathbb{E}[V_t - u_t \mid t, X_t] = 0$. The conditional variance of a two-point label is $\gamma_t(1-\gamma_t) \left\| v_T - v_{\mathrm{old}} \right\|^2$, proving the result. □

The analytic optimum contains the density responsibility γ_t, but training never evaluates it. Regression on samples from the joint law (B, X_t, V_t) learns the conditional mean implicitly. This is the standard conditional-flow-matching mechanism [1].

6.3 When the generated marginals are exactly the mixture

Suppose q_t and the flux in (83) are continuously differentiable, $q_t > 0$ on the relevant region, and u_t is continuous in time, Lipschitz in state, and has at most linear growth. Then the ODE

$$\dot{X}_t = u_t(X_t), \qquad X_0 \sim \rho_{\mathrm{base}}, \tag{92}$$

has a unique flow, and its density is the unique classical solution of (84). Therefore

$$(\Phi_t^u)_\# \rho_{\mathrm{base}} = q_t = (1-\alpha)\rho_{\mathrm{old},t} + \alpha\rho_{T,t}. \tag{93}$$

Under population optimization and sufficient model capacity, the square-loss minimizer equals u_t almost everywhere under $\omega(t)q_t(x)\mathrm{d}t\mathrm{d}x$. Arm A4 realizes the mixture path exactly when the fitted field admits a continuous/Lipschitz representative that equals u_t on the region traversed by the target flow; equality of abstract L^2 equivalence classes alone is not enough to define an ODE.

These regularity conditions are not automatic. Where both component densities are positive, differentiating (85) gives

$$\nabla\gamma_t = \gamma_t(1-\gamma_t)\left(\nabla \log \rho_{T,t} - \nabla \log \rho_{\mathrm{old},t}\right). \tag{94}$$

Low-overlap distributions can create a thin, stiff boundary where u_t switches between the two source fields.

6.4 Exact contraction statements

Proposition 6.3 (Contraction of the ideal marginal mixture)**.** *Assume the displayed KL terms are finite and both source laws have finite p-th moments. At every time t,*

$$\|q_t - \rho_{T,t}\|_1 = (1-\alpha)\,\|\rho_{\mathrm{old},t} - \rho_{T,t}\|_1\,, \tag{95}$$

$$\mathrm{KL}(q_t\|\rho_{T,t}) \le (1-\alpha)\mathrm{KL}(\rho_{\mathrm{old},t}\|\rho_{T,t}), \tag{96}$$

$$W_p^p(q_t,\rho_{T,t}) \le (1-\alpha)W_p^p(\rho_{\mathrm{old},t},\rho_{T,t}), \qquad p\ge 1. \tag{97}$$

The first line is equivalently an exact total-variation contraction. For $p=1$, (97) *is an equality; for* $p>1$ *it need not be.*

Proof. The signed density difference is $q_t - \rho_{T,t} = (1-\alpha)(\rho_{\mathrm{old},t} - \rho_{T,t})$, proving (95). Convexity of $r \mapsto r\log r$ gives

$$\left((1-\alpha)\frac{\rho_{\mathrm{old},t}}{\rho_{T,t}} + \alpha\right)\log\left((1-\alpha)\frac{\rho_{\mathrm{old},t}}{\rho_{T,t}} + \alpha\right) \le (1-\alpha)\frac{\rho_{\mathrm{old},t}}{\rho_{T,t}}\log\frac{\rho_{\mathrm{old},t}}{\rho_{T,t}},$$

which integrates against $\rho_{T,t}$ to (96). For Wasserstein distance, mix an optimal coupling between $\rho_{\mathrm{old},t}$ and $\rho_{T,t}$ with weight $1-\alpha$ and the identity coupling of $\rho_{T,t}$ with itself with weight α. Its transport cost is $(1-\alpha)W_p^p(\rho_{\mathrm{old},t},\rho_{T,t})$, proving the upper bound. The Kantorovich–Rubinstein dual formula gives equality at $p=1$. □

These are properties of the ideal mixture q_t. A finite-capacity student, if a global population optimum is attained, returns the best $L^2(q_t)$ fit to u_t and need not preserve them exactly. If the fitted field $\widehat{v}$ is L-Lipschitz, the same coupling argument as Proposition 1.2 gives the practical error bound

$$W_2^2(\widehat{\rho}_t, q_t) \le t\int_0^t e^{2L(t-s)}\mathbb{E}_{X_s\sim q_s}\|\widehat{v}(s,X_s) - u_s(X_s)\|^2\,\mathrm{d}s. \tag{98}$$

6.5 Cost and the branch-selection distinction

Exact A4 sampling requires states from both $\rho_{\mathrm{old},t}$ and $\rho_{T,t}$. In practice this means complete teacher trajectories for an α fraction of examples, or cached teacher trajectories reused across sampled training times. Evaluating the teacher field only on old states does not create samples from the teacher marginal and reduces A4 back to an old-occupancy field objective.

Choosing B once per trajectory gives the path-law mixture used in (87). Choosing independent branches at every small Euler step instead has conditional mean field

$$\bar{v} = (1-\alpha)v_{\mathrm{old}} + \alpha v_T, \tag{99}$$

and converges, as switching variance vanishes with the step size, to the arithmetic-average ODE of Arm A1. In general,

$$\bar{v} - u_t = (\alpha - \gamma_t)(v_T - v_{\mathrm{old}}), \tag{100}$$

so the two constructions coincide only in special cases such as equal marginals or equal fields.

Synthesis

Arm	mathematical object	exact statement	principal boundary
A0	old-occupancy field MSE	exact weighted regression / finite-capacity least-squares fit	no explicit trust region; no distribution objective
A1	quadratic field geometry	exact field barycenter and trust-ball projection; local Euler transport view	one on-policy step only scales A0; not a density mixture
A2	forward-target predictive risk	exact posterior only for a declared shared-covariance residual model	not a state-density ratio; calibration changes the surrogate
A3	stochastic one-step transition mixture	exact mixture-KL first derivative at the old policy	requires SDE covariance and $T_g = 1$; high-dimensional overlap collapse
A4	mixture of ODE time marginals	exact marginal path under population fitting and ODE regularity	requires teacher trajectories, capacity, and a regular mixture velocity

The five arms answer different questions rather than forming a single ranking. A0 is the cheapest ODE baseline. A1 is the appropriate deterministic trust-region baseline. A2 adds inexpensive, solver-agnostic evidence about teacher predictive competence. A3 is the strict transition-level probability-mixture construction when an SDE is acceptable. A4 is the strict deterministic probability-mixture comparator at the marginal level, paid for with teacher trajectories.

For practical ODE distillation, the main comparison should therefore keep these roles separate:

1. compare A0 and one-step A1 at a matched effective teacher-gradient scale, because Proposition 3.4 predicts equality otherwise;
2. test A1 with an actually enforced field RMS budget or multiple frozen-old inner updates;
3. compare A2 against an ungated target with the same timestep and norm normalization;
4. retain A3 as the stochastic local-mixture reference and log its joint/per-dimension gap and responsibility saturation;
5. use A4 on a smaller scale as the exact marginal-mixture comparator.

Main formal conclusion

There is no nontrivial deterministic continuation of the SDE transition-responsibility gate. What transfers to ODEs is the rescaled quadratic control geometry (A1), not the likelihood ratio. An exact deterministic probability mixture is still possible, but it lives at the level of ODE time marginals and is trained by two-source conditional flow matching (A4).

References

[1] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow Matching for Generative Modeling. *International Conference on Learning Representations*, 2023. https://arxiv.org/abs/2210.02747.

[2] Quanhao Li, Junqiu Yu, Kaixun Jiang, et al. DiffusionOPD: A Unified Perspective of On-Policy Distillation in Diffusion Models. 2026. https://arxiv.org/abs/2605.15055.

[3] Zhengpeng Xie, Li Lyna Zhang, Zeke Xie, and Mao Yang. Trust Region Policy Distillation. 2026. https://arxiv.org/abs/2607.04751.

[4] Bowen Ping, Xiangxin Zhou, Penghui Qi, Minnan Luo, Liefeng Bo, and Tianyu Pang. Flow-DPPO: Divergence Proximal Policy Optimization for Flow Matching Models. 2026. https://arxiv.org/abs/2606.11025.